

FOAM-BORN WORMHOLE PAIRS IN FULL NUMERICAL RELATIVITY: MERGER, SCATTERING AND COLLAPSE AS A RACE BETWEEN COLLISION AND DECAY

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I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale, both mouths in the same space [1–3]. More than one mechanism promotes them to macroscopic size: a Planck-scale Lorentzian wormhole caught in an inflating background is stretched with it and emerges macroscopic [4]; bubble and domain-wall cosmologies produce, without tuning, post-inflationary regions joined through wormholes [5, 6]; and a gas of relic wormholes has been run as a cosmology in its own right [7, 8]. What none of this work evolves is the aftermath: what a relic wormhole does once it is macroscopic and classical, and what two of them do to each other. That aftermath is this paper’s subject.

The classical theory fixes its two clocks. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable: perturbed one way the throat collapses to a black hole, the other way it inflates, on a timescale set by the throat radius [13, 14]. Every constituent destroys itself on its own schedule. Second, Geroch’s theorem forbids classical topology change [15, 16]: two wormholes cannot fuse into one, and none can be born. Birth is quantum [17], so a simulation takes over at $t = 0$ with the newborn state as its initial data—the stipulation every numerical-relativity binary makes, here with a physical story attached.

Together the two facts turn the standing objection to wormhole binaries into a selection rule. A binary has no astrophysical assembly channel—each throat dies faster than capture could pair two—but a foam-born pair needs none: creation and collision are the same event, at a birth separation of order the objects themselves. The race is collision against decay, and both rates belong to the classical solution. The scalar’s sign adds a second rule at no cost: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard routes [18–20]—or inflation. A foam population thus fans out into four fates—merge, scatter, collapse, inflate—across the three-parameter space of birth separation, perturbation and relative sign, and that space is what a family of evolu-

tions maps. The binary evolved here is the exterior approximation of a pair of handles (three asymptotic ends, the far sides passive); what merges is not two wormholes into one, which Geroch forbids, but two wormholes into one black hole with both throats inside it—the horizon does the work that topology change cannot.

The observable is not a chirp but a stochastic background from the merger epoch, redshifted to whichever band the mass scale puts it in [21–24]. The clean deliverable is therefore dimensionless and epoch-independent: the energy radiated per merger and its spectral shape in throat-mass units, which converts any bound on Ω_{GW} —including the nucleosynthesis cap that holds outside every detector band [25]—into a bound on the co-moving abundance of primordial wormhole mergers. Two caveats travel with the statement: the ghost scalar must exist at that epoch to hold the newborn throats open, and the merger’s scalar channel radiates negative energy whose cosmological consequence is genuinely open.

The paper is organised as follows. Sec. II sets out the model: the Einstein–phantom-scalar system, the drainhole, the binary superposition with its sign rule, and the declared seeds. Sec. III gives the numerical setup and the run matrix. The results then follow the campaign from one throat to two: the lone wormhole—fixed point, unstable mode, branching and collapse (Sec. IV); two throats at rest and the force between them (Sec. V); the head-on merger to a black hole (Sec. VI); the spiral merger and its capture boundary (Sec. VII); the gravitational waves of every channel—collapse, head-on, spiral, fly-by—gathered in one chapter with the vacuum twin (Sec. VIII); the numerical controls and systematics (Sec. IX). Sec. X confronts the rescaled waveforms with the LIGO–Virgo–KAGRA data products—template banks, ringdown, burst searches—and Sec. XI collects the findings. Every evolution runs on GPUs with `GRTeclyn` [36], the `AMReX`-based [37] port of `GRChombo` [38], continuing the campaign of Refs. [39, 40]. Throughout, $G = c = 1$.

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II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so $\rho < 0$ where the field's gradient dominates—the exotic stress that holds a throat open. With $V = 0$ this is exactly a ghost field. The sign flip lives entirely in the coupling to geometry: the scalar itself obeys the canonical wave equation $\square \phi = 0$, evolved first order in ϕ and $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the *lapse* rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no grid resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$ with areal radius $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2}$; the production throat $a = 2$, $m = 1$ has $R_{\min} = 3.8895$ at $r_t = 1.618$. The point $r \rightarrow 0$ is the far universe's spatial infinity compactified ($\chi \sim r^4$). The mass matters: at $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two of them do not attract at all.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so $\phi \rightarrow 0$ at the boundary, which is free for a massless field. Momentum enters through one Bowen–York term [30] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$. Since $\Pi = 0$ the matter momentum density vanishes and $\tilde{A}_{ij}$ is flat-space divergence free, so the momentum constraint is satisfied *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair's one discrete parameter: $\phi \rightarrow -\phi$ is exact for a single body, but a phantom mediator *repels* like scalar charges and *attracts* opposite ones, with $|F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 > 1$ always ($= 5$ for the production throat)—so like-signed pairs can never merge under their own gravity, and $\sigma = -1$ is the merger channel. A windowed one-body correction of the Helfer type [31] is implemented but off in production (Sec. IX A).

D. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi [1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (6)$$

which moves the throat's areal radius by $\approx 2\varepsilon$ with the sign of ε , and a quadrupolar factor

$$\psi \rightarrow \psi [1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (7)$$

about the z axis through the throat. K_{ij} and Π are untouched, so the momentum constraint stays exact. The spherical seed selects the branch; the quadrupolar seed has no $\ell = 0$ part—it gives a collapsing throat a quadrupole to shed—and with both on the cross term is $O(\varepsilon \varepsilon_2)$.

III. NUMERICAL SETUP

A. Evolution scheme

Evolutions use **GRTEclyn**: CCZ4 [32] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing [33] started from the exact static lapse $\alpha = e^u$ of Eq. (3), a Gamma-driver shift [34, 35] with $\eta = 1$ started from zero, fourth-order stencils, RK4, and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor; the bracket in Δt that makes it necessary is a control (Sec. IX A).

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ with the system at its centre; the base grid is 128^3 ($\Delta x = 0.5$). Refinement is by moving nested boxes centred on the tracked throats, shrinking by a factor of two per level (half-width 4 per throat at the production finest level) and re-drawn every 16 steps. Production runs use three refinement levels

(finest $\Delta x = 1/16$); resolution arms raise this to levels 4–7 where stated, and one control doubles the box to $L = 128$. Outer boundaries are Sommerfeld, backed by an absorbing sponge over $r = 24$ –32 that ramps extra dissipation up to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth into the layer; everything dynamical sits inside $r = 24$.

Each throat carries a compactified second infinity at its own centre, where $\chi \sim r^4$, so the evolved variables are flooded: $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant keeps χ unflooded down to 10^{-20} and applies the 10^{-8} floor only where $1/\chi$ is evaluated). Two further devices act only inside regions that are already trapped: an exponential damping of the scalar where the lapse has collapsed, and a smooth interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between an inner radius r_{full} and an outer r_{start} . All three are numerical devices, not physics; whether any of them is load-bearing is itself measured by twin runs (Secs. [VIB](#) and [IX A](#)).

C. Diagnostics

L^2 norms of the Hamiltonian and momentum constraints over the box are written continuously. Both throat centres are tracked, and about each centre the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat radius. On the same spheres the two null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*: a marginally outer trapped surface is an outermost zero of θ_+ with $\theta_- < 0$ (crossings at extrema of $R(r)$ are rejected), while a dynamic wormhole throat is marginally *anti-trapped* [\[29\]](#). Masses of such surfaces are Misner–Sharp [\[28\]](#), $2M/R = 1 + \frac{1}{4}R^2\theta_+\theta_-$; the scan uses coordinate spheres rather than a deformable finder, and quoted masses carry that shape systematic. Radiation is read from Ψ_4 interpolated onto spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms), decomposed in spin-weight -2 harmonics; all amplitudes are quoted as $r\Psi_4$. The in-code stream is cross-checked against an independent plotfile pipeline at the same radii. A per-cell autopsy reports the location and one-step history of any floating-point failure.

D. The campaign

The campaign is one-knob: every run differs from a named reference in a single setting, and every run has one line—its difference, its outcome, its caveats—in a registry shipped with the results pack (Sec. [XII](#)). Table [I](#) groups the 111 evolutions; the verdicts are the business of Secs. [IV](#)–[IX](#). An archived shakedown stage (12 of the 111) fixed the gauge, dissipation and tagging choices of Secs. [III A](#)–[III B](#) and is not used for physics.

TABLE I. The run matrix: 111 evolutions, grouped. Every group varies one knob against a named reference; per-run lines live in the shipped registry. Body parameters are $a = 2$, $m = 1$ ($R_{\text{min}} = 3.89$), $\sigma = -1$ and $d = 12$ unless stated.

group	knob(s)	cells	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; level 4	8	IV C
quadrupolar kicks	$\varepsilon_2 = 0.01, 0.05$; levels 3–4	4	VIII A
two throats at rest	σ ; $a = 1$ –3; $d = 12$ –18	9	V
placement probes	$d = 6$ –48, one step each	18	V C
head-on, $d = 8$	fill; levels 3/5; down-step	8	VI
half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VIC
spiral, $p = 0.12$	levels; N ; gauge; floor; fill	29	VII
capture scan	$p = 0.20$ –0.45	4	VII A
Helper twins	correction; window; levels	6	IX A
vacuum BBH	same d , p , masses; no scalar	2	VIII E
shakedown (archived)	gauge, dissipation, tagging	12	III A

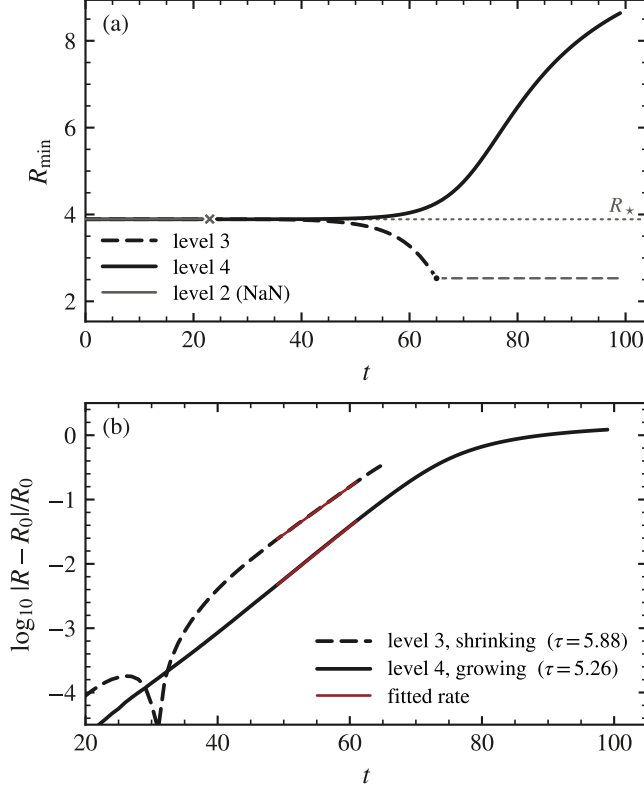


FIG. 1. The undeclared seed: one exact throat, two resolutions. (a) Minimal areal radius of the lone drainhole evolved from exact static data at refinement levels 2–4. The exact solution is a fixed point (R_* , dotted); each level holds it until its own truncation error excites the unstable mode, and the fate flips with the grid: level 3 (dashed) collapses, level 4 (solid) inflates. Level 2 (grey) dies of an origin failure at $t = 24.2$ (cross) with the throat still exact to 0.12%—too coarse to reach the branch at all. Dot and muted line as in Fig. 2: the scan loses the level-3 throat at $t = 65$ while the run continues to $t = 100$ behind a marginally trapped surface (areal radius $3.23 \rightarrow 2.37$). (b) The deviation grows exponentially on both grids: plateau fits over $t = 49$ – 61 give e-fold times $\tau = 5.88$ (level 3) and 5.26 (level 4)—the same unstable mode to 9%, $T = \tau \alpha_{\text{th}}/R_* = 0.87$ and 0.78 against the González-Guzmán-Sarbach band 0.68 – 0.76 [14]—with opposite sign. The fate is the truncation seed’s; the mode is the throat’s.

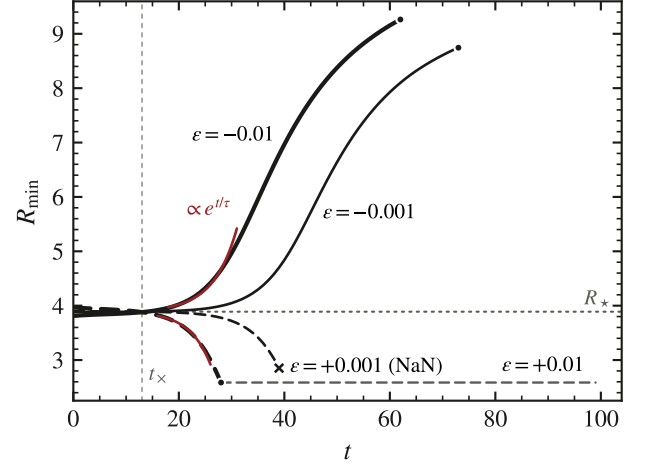


FIG. 2. Declared-seed branching of the lone throat (level 3). Areal throat radius of the drainhole ($a = 2$, $m = 1$; $R_* = 3.8895$, dotted) under the kick of Eq. (6): $\epsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light); dashed = outward ($\epsilon > 0$). The fate is opposite to the push, and each pair crosses itself at the ϵ -independent $t_{\times} \simeq 13.0$ —linear response. Burgundy: exponential fits $|R - R_*| \propto e^{t/\tau}$, $\tau = 4.5$ ($t = 18$ – 31) and 4.0 ($t = 16$ – 26)—steeper than the unkicked mode’s 5.88 (Fig. 1) while the kick’s transient decays. Dots: the scan loses the throat (the inflating runs continue to $t = 100$, still growing); cross: death at $t = 40.1$. The $\epsilon = +10^{-2}$ arm holds a horizon from $t = 11$ (areal radius $3.88 \rightarrow 2.34 \rightarrow 2.57$) and runs clean to $t = 100$; the muted line carries its last reading. A tenfold smaller kick arrives 8–12 units later.

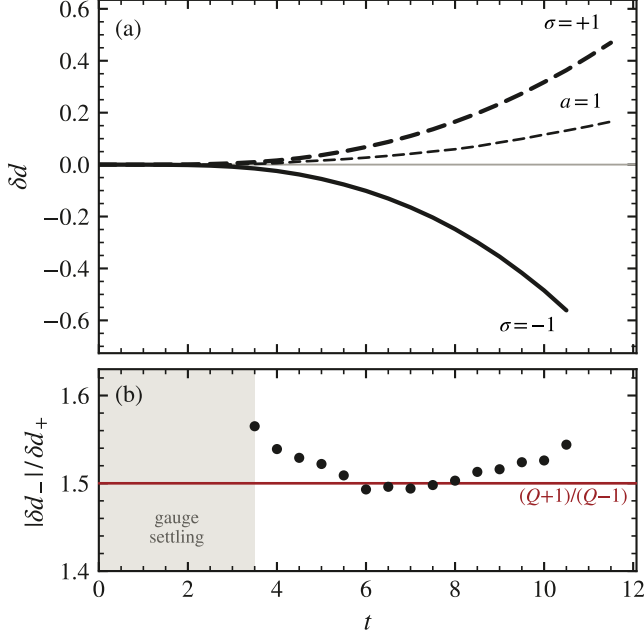


FIG. 3. Attraction and repulsion switched by one sign ($d = 12$, released from rest). (a) Separation change of three pairs one knob apart: the like-oriented pair ($\sigma = +1$, dashed) pushes apart; flipping one field ($\sigma = -1$, solid) turns the push into a pull; the narrow like pair ($a = 1$, light) opens $2.9\times$ more slowly—the force follows the throat width (the scalar charge), not the mass, which is equal in all three. Separations are inverse- χ -weighted centroids of the midplane χ pits, sub-cell where the mouth tracker’s 0.03 grid quantum would swamp the early motion. (b) The flipped pair’s infall over the like pair’s escape: with $Q \equiv |F_\phi/F_{\text{grav}}| = 5$ (Sec. II C), pull and push are $(Q+1)$ and $(Q-1)$, predicting $3/2$ (burgundy); measured 1.52 ± 0.02 over $t = 3.5\text{--}10.5$. Coordinates under-read the motion while the gauge settles (band), but that systematic is common to both arms and cancels in the ratio: the two runs differ in exactly one parameter.

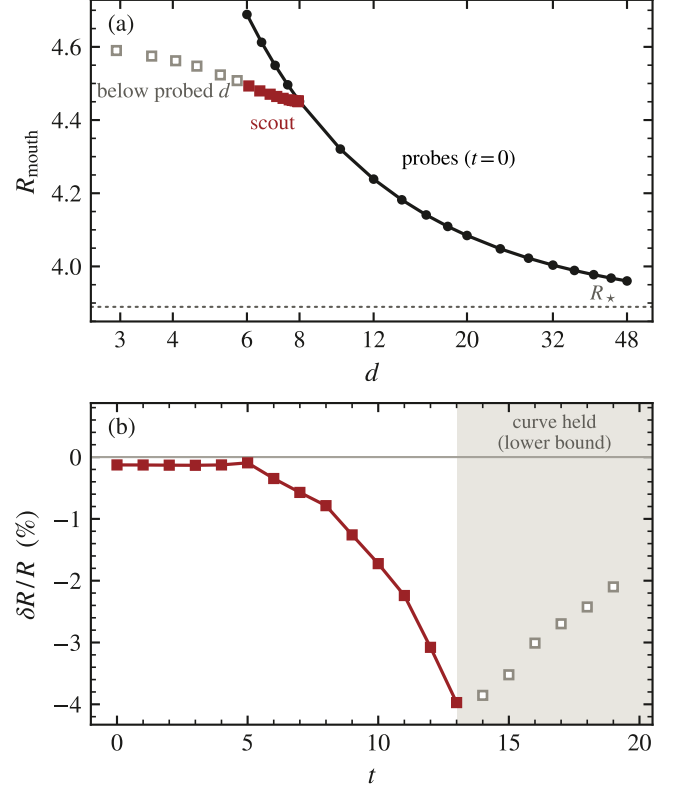


FIG. 4. Placement is on the ruler; the squeeze is not. (a) Per-mouth areal radius of two exact drainhole throats placed at rest, read at $t = 0$ by one-step probes at separations $d = 6\text{--}48$ (ink): the pair reads wider than the isolated throat (R_* , dotted), the excess falling as $d^{-1.16}$ —the neighbour’s field on the ruler, not a dynamical widening. Burgundy squares: the head-on scout’s mouths (the $\sigma = -1$ pair released from rest at $d = 8$, Sec. VI) at the separation it had reached; open squares, here and in (b): the scout closer than the closest probe. (b) The scout against the curve at its own separation, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$. The -0.13% plateau to $t = 5$ is the mouth tracker’s grid snap (d reads 7.94 for a true 8) and calibrates the systematic; the throats’ own squeeze then grows to -4.0% by $t = 13$ ($d = 6.1$). In the band the curve is held at its last point, so the magnitude is a lower bound—the apparent recovery is the ruler’s, not the throats’. A boost can only raise a coordinate-sphere reading, so a reading below the curve is a real squeeze, if anything underestimated.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

B. The unstable mode

C. Branching: collapse versus inflation

D. The collapse endpoint

V. TWO THROATS AT REST

A. The sign rule

B. The force law

C. Placement systematics

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

B. Through the wall to a black hole

C. The half-mass pair

VII. THE SPIRAL MERGER

A. Capture and scattering

B. Inspiral and collapse

C. The wall

VIII. GRAVITATIONAL WAVES

A. The collapsing throat

B. The head-on ringdown

C. The spiral's carried burst

D. The fly-by burst

E. The vacuum twin

IX. CONTROLS AND SYSTEMATICS

A. Numerical systematics

X. PATTERN MATCHING AGAINST DETECTOR DATA

The simulated waveforms are scale-free: with the throat mass M as the unit, $r\Psi_4$ rescales to detector strain by $h \propto M/D$ and $f \propto 1/M$, so one numerical waveform covers every candidate mass at once. This section confronts the rescaled $(2,2)$ and $(2,0)$ waveforms with the LIGO–Virgo–KAGRA data products on three fronts: matched filtering [26], the ringdown, and the unmodelled searches.

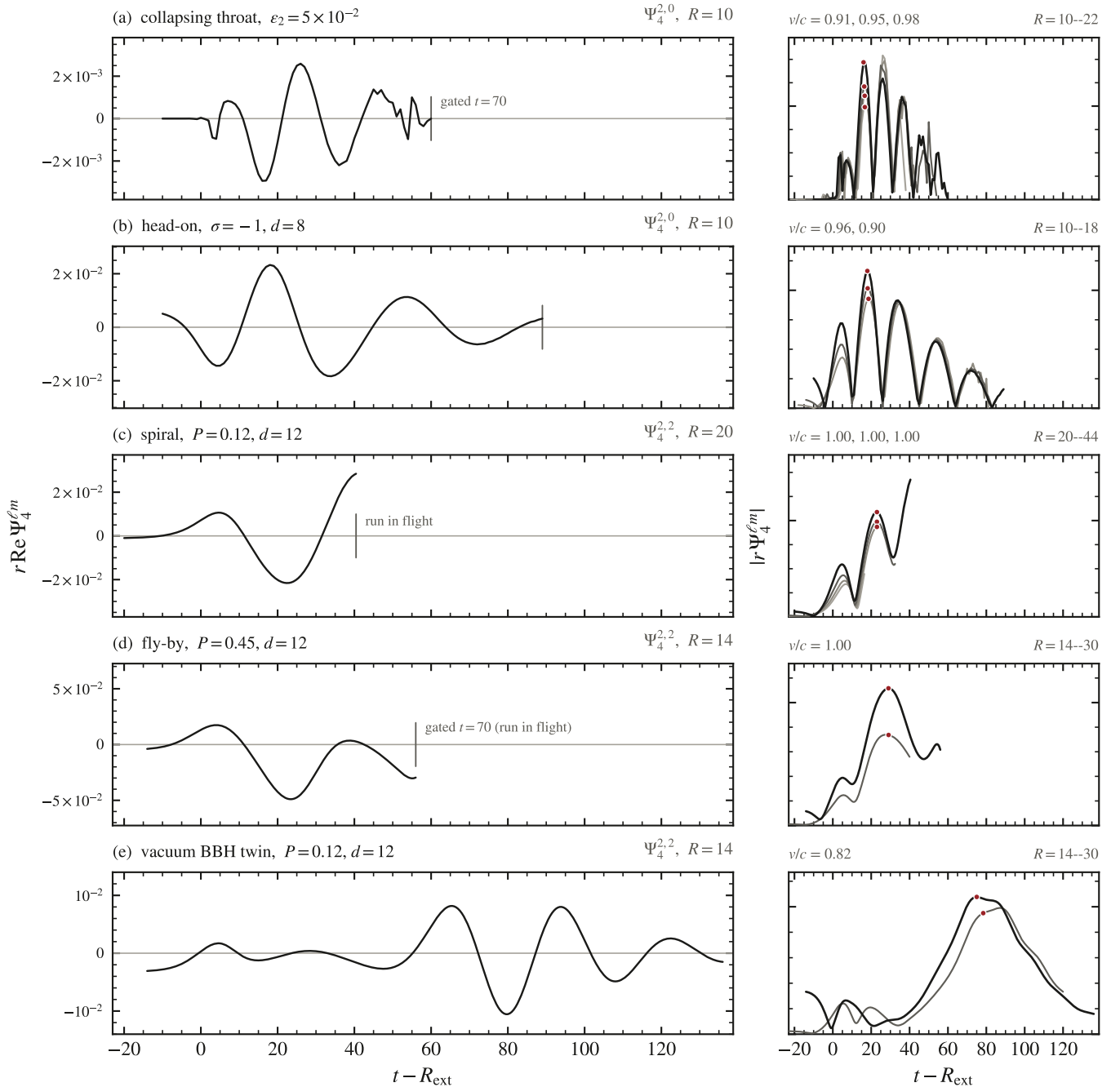


FIG. 5. The waveform gallery: every channel's radiation, one row per scenario, one retarded clock. Left column: $r \text{Re} \Psi_4$ of the dominant mode at the innermost extraction sphere, each row on its own stated scale—the lone throat radiates an order of magnitude below the mergers, and the fly-by is the loudest channel of all. Half-height caps mark record ends: the collapsing throat's stream is junk-gated at $t = 70$, the spiral record ends at $t = 60$ with the run still in flight, and the fly-by is gated at $t = 70$, where the core's late disturbance reaches the $R = 14$ sphere (its close-out runs continue). Right column: the envelope $|r \Psi_4^{\ell m}|$ at every extraction sphere on its own retarded time (ink to faint, inner to outer): a signal travelling at c and falling as $1/r$ folds onto one curve. Burgundy dots mark the correlated wavefront, and the quoted v/c is the lag of the whole complex waveform between neighbouring spheres over their common retarded window—peak-to-peak timing fails on flat-topped envelopes (the vacuum twin's merger plateau is ~ 15 units wide at both spheres, and its peak times alone would read $v = 0.57c$ where the waveform lag reads $0.82c$). Measured speeds: throat $0.91/0.95/0.98$, head-on $0.96/0.90$, spiral $1.00/1.00/1.00$, fly-by 1.00 , vacuum twin 0.82 . Every channel propagates at c within the output cadence: radiation, not a gauge or constraint mode.

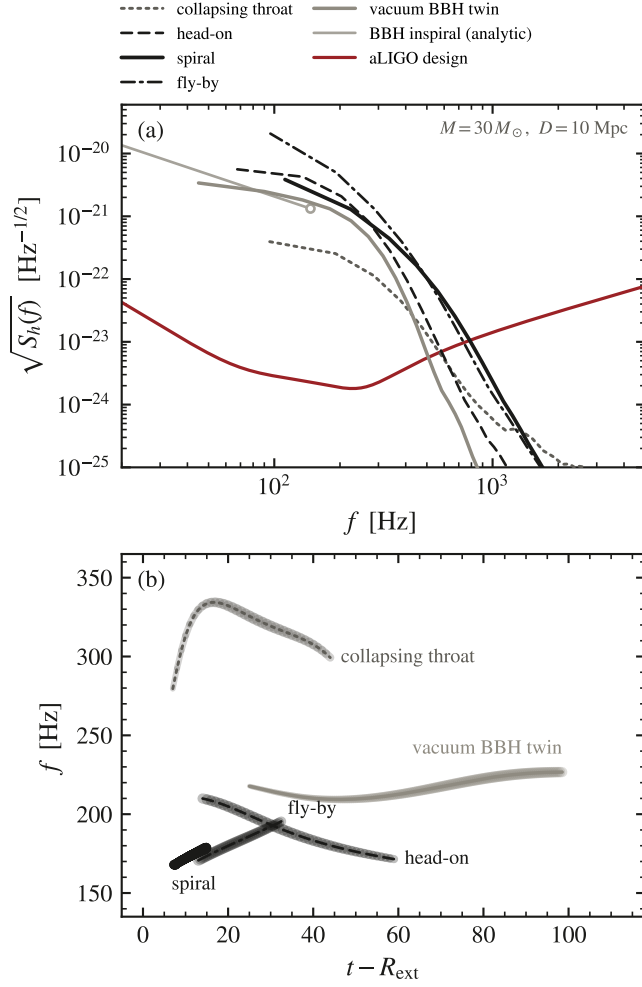


FIG. 6. Every source on a detector, at one shared calibration: total mass $M = 30 M_\odot$, distance $D = 10 \text{ Mpc}$; strain scales as M/D and frequency as $1/M$. (a) Strain amplitude spectral densities of the innermost-sphere records over the aLIGO design floor (burgundy). Every channel clears the floor around 100 Hz: the fly-by peaks at $2.1 \times 10^{-20} \text{ Hz}^{-1/2}$ (96 Hz); the head-on, spiral and vacuum twin lie within a factor of two of one another ($3.4\text{--}5.6 \times 10^{-21}$); the lone collapsing throat sits an order below (3.9×10^{-22}). The faint straight line is the analytic Newtonian inspiral of the same equal-mass binary, $|\tilde{h}(f)| \propto \mathcal{M}^{5/6} f^{-7/6}$, drawn to its ISCO at 147 Hz through the same normalisation as the numerical record: the ramp an astrophysical binary rides into the band where the simulated signal begins. (b) Peak-frequency tracks of the Morlet wavelet transform of each record (the line's width is the signal's own envelope; tracks are drawn only above 0.2 of each record's peak and outside the cone of influence). The shapes separate the sources: the collapsing throat rings in an arch at 280–334 Hz, the head-on bell slows from 210 to 170 Hz, the vacuum twin climbs through merger onto a 227 Hz ringdown shelf, and the spiral and fly-by (records in flight) enter as rising arcs at 165–195 Hz.

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- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064 (2016).
 - [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
 - [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
 - [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).
 - [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
 - [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
 - [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
 - [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
 - [13] H. Shinkai and S. A. Hayward, “Fate of the first traversable wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
 - [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009); see also **26**, 015010 (2009).
 - [15] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
 - [16] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
 - [17] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
 - [18] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
 - [19] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
 - [20] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
 - [21] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).
 - [22] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
 - [23] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
 - [24] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
 - [25] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
 - [26] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
 - [27] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
 - [28] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
 - [29] D. Hochberg and M. Visser, “Dynamic wormholes, anti-trapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
 - [30] J. M. Bowen and J. W. York, “Time-asymmetric initial data for black holes and black-hole collisions,” *Phys. Rev. D* **21**, 2047 (1980).
 - [31] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, “Malaise and remedy of binary boson-star initial data,” *Class. Quantum Grav.* **39**, 074001 (2022).
 - [32] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
 - [33] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
 - [34] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
 - [35] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
 - [36] GRTL Collaboration, *GRTEclyn*, <https://github.com/GRTLCollaboration/GRTEclyn>.
 - [37] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).
 - [38] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
 - [39] N. M. Shirokov, “Wormhole dynamics: nonlinear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
 - [40] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive-negative mass binary,” submitted (2026).